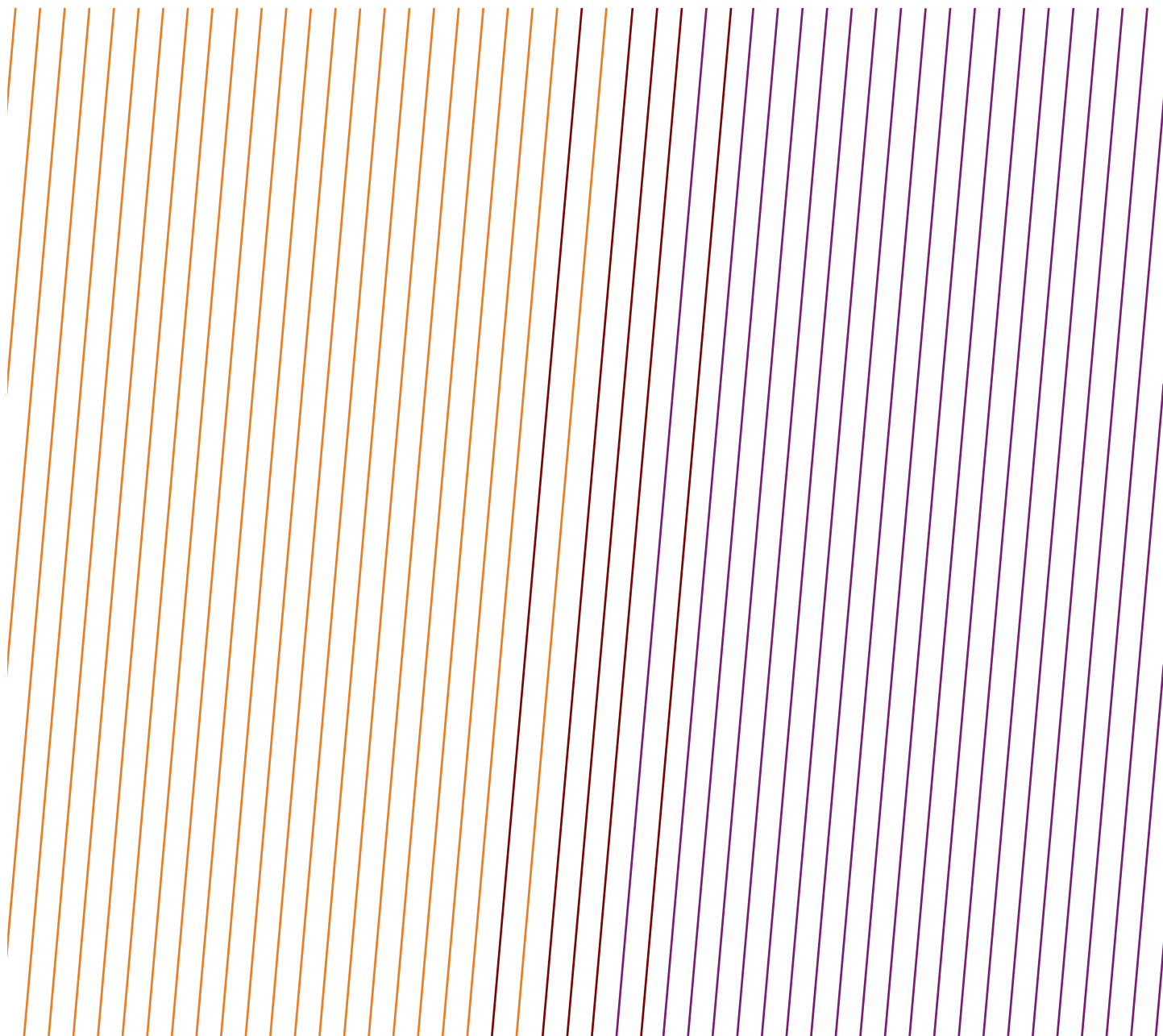


DATA SERIES

Safety performance indicators – 2022 data – Fatal incident reports



Acknowledgements

IOGP thanks those companies that have participated in the data collection programme.

This report was produced by the Safety Committee.

Feedback

IOGP welcomes feedback on our reports: publications@iogp.org

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DATA SERIES

Safety performance indicators – 2022 data – Fatal incident reports

Revision history

VERSION	DATE	AMENDMENTS
1.0	June 2023	First release

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AFRICA ONSHORE

DATE: 01 Aug 2022

COUNTRY: Nigeria

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Maintenance, inspection, testing

RULE: Hot work

Employer: Contractor; **Occupation:** Maintenance, Craftsman; **Body part:** Leg/knee/hip; **Nature of injury:** Burn

NARRATIVE:

During hot work activity, there was fire incident which impacted two contractor personnel. The more severely injured person was transferred to a specialist burn centre, but he unfortunately passed away.

WHAT WENT WRONG:

Contractor personnel were performing planned hot work inside a tank on three stilling wells. These works were required to prepare the tank for calibration (hydrotesting) prior to returning the tank to service. IP1 was grinding datum plate bolts off stilling well 2. IP2 was welding a datum plate on the base of stilling well 3, approximately 1.7m away from stilling well 2. After removing two of the four bolts from stilling well 2, crude oil that had accumulated in its 2" compartment began flowing from stilling well 2, spraying in the direction of stilling well 3. Because of the welding or grinding work, the released crude oil ignited. Both IP1 and IP2 sustained burns in the resulting fire.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Tank certification as hydrocarbon free – tanks shall be certified as 'gas free' after tank draining and cleaning, in fact, free of residual hydrocarbons. Personnel certifying tanks shall be aware of all potential locations of residual hydrocarbons, including those where use of standard gas detector would be unlikely to detect their presence.

- Tank cleaning – Cleaned tanks shall be free of residual hydrocarbons. Tank cleaning personnel shall have 'front of mind' awareness of potential locations of residual hydrocarbons.
- Emergency notification – Emergency service personnel shall promptly and successfully notify in case of emergencies. A consistent (across assets/locations) and simple process shall be in place for company staff and contractor personnel to provide emergency notification.
- Personal protective equipment (PPE) – Contractor personnel shall understand and fully comply with the PPE Manual.
- DEP standard drawing for a stilling well – Company and contractor personnel should be skilled enough to understand and correctly interpret the drawing content.

CAUSAL FACTORS:

No Causal Factors Allocated

AFRICA OFFSHORE

DATE: 05 Jul 2022

COUNTRY: Egypt

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Falls from height

ACTIVITY: Drilling, workover, well operations

RULE: Working at height

Employer: Contractor; **Occupation:** Other; **Body part:** Neck/torso/spine; **Nature of injury:** Crushing

NARRATIVE:

A floorman was working at a BOP maintenance area at jack up cellar deck at a height of almost 17 meters from the platform top deck. After assembling the BOP shear rams, some housekeeping activities (including tools recovery and collection) were being performed by a crew of total six members. At the time of the incident, the IP was standing on the walking area of the BOP nearby the Coflex (right side of BOP). He had previously received some tools from height in order to recover them (at least a metal plate and a chain block, with total weight of approximately 15 kg) from one crew member standing on the left side of BOP, third floor. The IP approached the right side of the area, through the walking area around the BOP – passing in the rear side of BOP itself. Most likely his intention was to approach the BOP first floor from the right side to collect other tools or to place the received ones in a dedicated area, before the final collection. In any case he put his foot above the Coflex hose connected with the BOP (inherently slippery) and fell down through an open area of almost 1 metre (between the tubular handrail and the walking area around the BOP) from a height of 17 metres. The falling dynamic is compatible with both hypothesis: either the IP lost his balance due to the weight of the tools (approximately 15 kg) or while approaching the BOP first floor. A medevac was immediately activated but the IP was declared dead in the hospital.

WHAT WENT WRONG:

- Poor planning activities and role assignment: the IP started working on the POB area as part of BOP maintenance crew but then was sent to the shale shaker area. After finishing the activity there, the IP went back to the BOP area to recover his original task.
- Complacency and training/competences gaps: the IP and crew members demonstrated poor risk perception and awareness on working at height.
- Poor Risk Assessment: Job Safety Analysis was attached to the PTW but risks related to fall from height, manual handling, presence of slippery areas and absence of barriers were not included in the risk analysis.
- Incomplete operating instruction: Job Safety Analysis reported the list of main tasks for BOP maintenance (including method of statement) but this list did not include all the risks related to working at height. JSA did not include tasks and risks related to the lifting operations that were part of the activity and a dedicated permit to work for such activities was not issued.
- Tool Box Talk (TBT) not effective: TBT performed but crew did not flag the falling or slipping/tripping hazards and risks related to improper manual handling.
- PPE not used: the IP and some of the crew members not wearing the personnel fall protection equipment foreseen by the PTW.
- Work environment: presence of an open side in the BOP area of around 93 cm height between the first handrail and the floor. Such free space is wide enough to allow a person to fall through it. No other barriers and signs were present.

AFRICA OFFSHORE

- Design specifications not respected: the top and mid rails of the handrail were installed at a height higher than as it shall be as per standards.
- MOC: installation of tubular handrails at height different than as per standard, not analyzed or risk assessed through MOC process.
- Lack of supervision: supervision not effective and previous inspections did not identify the unsafe conditions of the working area.
- Poor Housekeeping: the hose that the IP stepped on was wet with oil and grease and thus slippery. The hose was beside the walkway area without segregation or barrier.
- Contractor Management: opportunities for improvement have been found in the contract management process, as new higher HSE requirements were included in the contract but only inspection for rig acceptance was done when hiring the Rig.
- Emergency Management & Response: delays in helicopter take-off due to some equipment to be loaded, medevac procedure not properly followed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- All the risks related to Work at Height shall be evaluated (including tools handling), assessed in the Permit to Work and deeply discussed during the pre-job meetings, briefing and Tool Box Talk.
- Ensure all personnel involved in the operation are aware of the operating instructions and methods of statement.
- Refresh PTW procedure (including Tool Box Talk and Job Safety Analysis philosophy) and risks related to routine and non-routine activities. Ensure all personnel involved are aware of all the activity-related risks, duties & responsibilities, and safety precautions.
- Ensure opening of a dedicated PTW / Risk Assessment / Lifting Plan for any Lifting Operation requiring it.
- Ensure adequate supervision of activities.
- The worksite area shall be tidy before beginning a task, at the end of shifts and on completion of the task.
- All the Rig operators need to be aware of all the risks related to the activities.
- In cooperation with a third party, assess the competences, duties, rig organization to understand if sufficient, skilled manpower is available at the rig to perform the activities, as per duties and planning. Ensure periodic follow up of 3rd party inspections/assessments.
- Ensure proper Contractor Management by implementing specific safety programs and initiatives and verifying contractor compliance with contract requirements (eventually with support of a third party verification).
- Provide barriers to prevent fall risk areas as much as possible, and safety signs to make the crews aware of the risk of falling and the presence of slippery areas and area segregations / no go zone.
- Clearly assign roles & responsibilities in the medevac procedure; periodically refresh all Rig Personnel about the Communication Flow Chart in case of medevac; perform regular medevac Drills without previous announcement/notification.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

ASIA/AUSTRALASIA ONSHORE

DATE: 16 Mar 2022

COUNTRY: China

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

Employer: Contractor; **Occupation:** Drilling/Well Servicing Operator; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Loss of consciousness

NARRATIVE:

During a drilling operation, began to connect drill pipe columns. After the second column was connected and the hook rose to the second floor of the derrick, the driller moved the column to the drill pipe box for operation. When the derrick man was ready to place the column but had not opened the elevator valve, the traveling block hook suddenly dropped, and the elevator fell off and fell to the drill floor, hitting the driller who was working in the wellhead area. His safety helmet was dropped, and the IP fell unconscious on the spot. The operation was immediately stopped and first aid was carried out. The injury was fatal.

WHAT WENT WRONG:

During the process of connecting the drill pipe column of the well, the drilling elevator loosened and fell, hitting the floorman on the drill floor.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

In order to effectively strengthen the company's safety production work, improve the safety production responsibility and safety production awareness of all employees, and prevent the recurrence of production safety accidents, a major safety inspection was carried out throughout the company in strict accordance with the principle of "four not to let go". At the same time, contractor management was standardized and strengthened in accordance with the main responsibility of enterprise safety production, strengthened ideological awareness, and standardized and strengthened the contractor management. The rectification report has been formed by strengthening the implementation of the system, eliminating the "double skin", systematically carrying out the construction of the drilling and completion supervision team, improving the on-site emergency response ability and rescue ability, and implementing the requirements of various safety production work in 2022.

1. Strengthen the organization to watch the safety promotional film to ensure the full coverage of learning.
2. Accident case study warning.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

ASIA/AUSTRALASIA ONSHORE

DATE: 08 Oct 2022

COUNTRY: Pakistan

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Maintenance, inspection, testing

RULE: Work authorization

Employer: Contractor; **Occupation:** Drilling/Well Servicing Operator; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Crushing

NARRATIVE:

During Preventive Maintenance (PM) of emergency shutdown valve at a wellsite, the actuator of the valve abruptly burst and its metallic part hit a nearby contractor. The injured Person (IP) was taken to the clinic, where he passed away. The actuator detached parts hit the flow line Pressure Differential Transmitter (PDT) tubings, which resulted in minor crack on a Down Stream (D/S) welded nipple. This led to gas leakage.

WHAT WENT WRONG:

- Risk Assessment not revised following the deviation from Scope of Work (SOW).
- Tight schedule, only one Instrument team was on site to perform all the planned activities.
- Complacency due to routine activity being perceived as low risk, leading to overconfidence of workers.
- The Instrument Technicians themselves deviated from SOW without applying Management of Change process.
- Stop Work Authority not applied.
- Ineffective supervision. Supervisors were present on site, but focused on other activities.
- Lack of means of communication. Unavailability of radio device for communication between control room and work area leading to pressurized actuator for more than 7 minutes.
- Lack of awareness of performing authority related to process hazards leading to risk underestimation.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Any change or deviation to design, asset, process, work procedure or practices shall be accurately identified, reported and assessed by competent and trained personnel.
- The job shall be suspended in case of change in the SOW, and Risk assessment revised to involve all the stakeholders.
- Ensure all personnel involved in the operation are aware of the operating instructions and methods of statement.
- Ensure proper communication among working team members on site.
- Ensure competence, awareness and training of the personnel involved in operations.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

ASIA/AUSTRALASIA ONSHORE

DATE: 26 Jan 2022

COUNTRY: Pakistan

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Assault or violent act

ACTIVITY: Production operations

RULE: Other

Employer: Contractor; **Occupation:** Other; **Body part:** Neck/torso/spine; **Nature of injury:** Shot

NARRATIVE:

On 26th January 2022, a contractor security guard was killed during a terrorist attack. Some parts of the facility were also damaged by the terrorist.

WHAT WENT WRONG:

Terror attack.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Security procedure was reviewed.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 20 Jan 2022

COUNTRY: Pakistan

NUMBER OF DEATHS: 1

FUNCTION: Construction

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Transport - Land

RULE: Driving

Employer: Contractor; **Occupation:** Transportation Operator; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Crushing

NARRATIVE:

A Contractor's water tanker rolled over while it was travelling towards a right of way from a water point. There were two passengers in the tanker; a driver and a local person. Both were seriously injured during the accident. Medical treatment was provided to the local person at the hospital. The local person had been engaged by the deceased driver on the day of accident. The driver passed away while he was being transferred to hospital by ambulance.

WHAT WENT WRONG:

The driver did not go to a lower gear on the flat surface before the slope. Instead, he attempted to lower the gear while descending the slope which resulted in continuous momentum gain. Due to extreme panic, sudden uncontrollable fear, loss of nerves, and his complete attention towards changing the gear, the driver failed to apply any of the brakes, neither service brakes nor hand brakes. This can be also confirmed by the fact that no brake marks were visible on the right of way where the accident happened.

ASIA/AUSTRALASIA ONSHORE

Rough terrain, inclined profile of land and filled water in the bowser contributed in increasing momentum of the descending bowser. Driver did not follow the speed limit, he also did not follow the safe driving practice required for the hilly terrain.

Failure to effectively implement required actions for controlling the hazard. The driver tried to direct the bowser towards crown/shoulder of the road to stop the vehicle, which instead lifted the tanker and ultimately it rolled over. Supervisor failed to ensure that every worker should attend TBT before the job begins. Helper was appointed by driver without any prior information or approval. Therefore, helper was missing the basic skill set and orientation required to perform the job.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Company Driving Policy and Safe Driving Practices shall always be followed.
- Vehicle's speed shall always be reduced, and gears should be lowered before descending on slope.
- Speed Limit shall always be followed.
- Driving Techniques shall always be adapted as per the Terrain and surrounding conditions.
- Attendance of all workers shall be ensured in ToolBox Talks (TBTs).
- Hazards associated with workplace and surrounding conditions must be discussed and communicated to workers in all TBTs.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: 05 Mar 2022

COUNTRY: Thailand

NUMBER OF DEATHS: 1

FUNCTION: Unspecified

CAUSE: Struck by (not dropped object)

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

Employer: Contractor; **Occupation:** Unknown; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Cut, Puncture, Scrape

NARRATIVE:

There was an explosive incident while conducting an under-rig-floor slickline operation. The explosion resulted in one fatality and two injured persons. All of the affected personnel work for contracted companies.

WHAT WENT WRONG:

Perforation gun was unintentionally activated.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not yet available.

ASIA/AUSTRALASIA ONSHORE

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

ASIA/AUSTRALASIA OFFSHORE

DATE: 05 Jul 2022

COUNTRY: Malaysia

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Water related, drowning

ACTIVITY: Diving (incl. decompression), subsea, ROV

RULE: Other

Employer: Contractor: **Occupation:** Unknown: **Body part:** Respiratory system: **Nature of injury:** Unspecified

NARRATIVE:

Two divers were involved in an incident during underwater hull repair project for a Floating Production and Storage & Offloading (FPSO) vessel at an offshore production field. The work was conducted at seat with depth of 6 meters (approximately 20 feet). The divers left a Diving Support Vessel (DSV) to perform an adapter mating test. While positioning the adapter in its intended location, the adapter suddenly lost its buoyancy and sank. One of the divers (Diver 1) was seen sinking together with the adapter to a depth of 55 meters (approximately 180 feet), entangled with the adapter rope. Diver 2 rescued Diver 1 and swam up together with unconscious Diver 1 to the surface. Diver 1 was taken to DSV compression chamber, but had lost vital signs at that point. He was pronounced dead four hours after CPR was attempted to revive him.

WHAT WENT WRONG:

Deviation from approved procedure (Diving Safety Management System Manual) during diving operation and emergency procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Existing Company Site Representatives (CSR), offshore diving supervisors and superintendent shall complete trainings as per IMCA Client Offshore and IMCA C 003.
- Future SISO involving diving operations to include company standards in Diving Operations Management.

CAUSAL FACTORS:

No Causal Factors Allocated

ASIA/AUSTRALASIA OFFSHORE

DATE: 03 Mar 2022

COUNTRY: Malaysia

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

RULE: Hot work

Employer: Contractor; **Occupation:** Unknown; **Body part:** Arm/elbow/shoulder; **Nature of injury:** Unspecified

NARRATIVE:

A fire incident occurred at an offshore production platform during hot work activity inside a pressurized habitat for deck plate replacement near a Booster Compressor Suction Scrubber involving contractor personnel. The fire incident has resulted in 1 one fatality after 3 three days of treatment at a burn intensive care unit, and 1 one burn injury.

WHAT WENT WRONG:

Hot work activity carried out inside habitat on live HC piping.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Thorough risk assessment for any hot work activity during live plant environment.
- Avoid erecting the habitat on the live HC line.

CAUSAL FACTORS:

No Causal Factors Allocated

EUROPE ONSHORE

No fatal incidents reported

EUROPE OFFSHORE

No fatal incidents reported

MIDDLE EAST ONSHORE

DATE: 11 Oct 2022

COUNTRY: Bahrain

NUMBER OF DEATHS: 1

FUNCTION: Unspecified

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Line of fire

Employer: Contractor; **Occupation:** Drilling/Well Servicing Operator; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Crushing

NARRATIVE:

During lifting activity on rig site, a TGB joint box disconnected while lifting a pipe that was not centred properly. The TBG joint box suddenly disconnected from the saver sub pin and fell on the work floor. The TBG joint struck the Subcontractor Personnel's head, causing a fatal injury.

WHAT WENT WRONG:

- SOP require further detail enough and missed second checker.
- SPAC not detailed enough and ambiguous.
- Not enough supervision.
- JSA not detailed enough.
- Lights need improvement.
- Late communication (Subcontractor was standing in the exclusion zone assuming that the joint was connected).

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Proper connection should be made before hoisting the pipe with the power swivel.
- All non-essential personnel should stay outside the safety exclusion zone during lifting operations including while lifting the work joints using the power swivel.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: 31 Oct 2022

COUNTRY: Oman

NUMBER OF DEATHS: 1

FUNCTION: Unspecified

CAUSE: Struck by (not dropped object)

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

Employer: Contractor; **Occupation:** Manual Labourer; **Body part:** Internal organs; **Nature of injury:** Crushing

NARRATIVE:

During pre-move operations, the IP (a banksman) was caught between an oilfield truck and a tool basket. The IP was initially treated on site by the rig medic, then transferred to the clinic, stabilized, and transferred to Hospital. Incident investigation on going. Driver D&A results were negative.

MIDDLE EAST ONSHORE

WHAT WENT WRONG:

The spotter was crushed by the moving truck.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Vehicle must not move if spotter not visible. Back up cameras and proximity sensors should be utilized.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 27 Jan 2022

COUNTRY: Oman

NUMBER OF DEATHS: 2

FUNCTION: Unspecified

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

RULE: Hot work

Employer: Contractor; **Occupation:** Maintenance, Craftsman; **Body part:** Internal organs; **Nature of injury:** Burn

Employer: Contractor; **Occupation:** Maintenance, Craftsman; **Body part:** Internal organs; **Nature of injury:** Burn

NARRATIVE:

A fire occurred at a gas plant heater. Emergency Response, Support and Security teams dispatched immediately to the incident location. The incident resulted in 3 injured persons.

WHAT WENT WRONG:

Flash fire ignited by workers. Leak caused when removing insulation. The self-tapping screws to hold insulation had unknowingly penetrated the vessel.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

MIDDLE EAST ONSHORE

DATE: 15 Jul 2022

COUNTRY: UAE

NUMBER OF DEATHS: 2

FUNCTION: Drilling

CAUSE: Falls from height

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

Employer: Company; **Occupation:** Drilling/Well Servicing Operator; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Unspecified

Employer: Company; **Occupation:** Engineer, Scientist, Technician; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Unspecified

NARRATIVE:

During rig move and rigging down operations, the drilling crew was dismantling the TDS. The task was to disconnect the TDS from the guide beam by disengaging the tie-back hook pins.

Two Floormen were assigned to remove the upper tie-back hook pins while the Derrickman was to remove the lower tie-back hook pins. Three of the four tie-back hook pins were successfully disengaged, and when the Derrickman was detaching the last tie-back hook pin, the TDS skid swung towards the Floorman's (IP 1) position and struck him against the mast beam. IP # 1 sustained fatal chest injuries.

Simultaneously, at the rig floor, the Assistant Electrician (IP 2) was working with the Instrumentation Technician to remove and lower the disconnected TDS service loop cables from the rig floor to the ground. While the Instrumentation Technician pushed the cable away from the handrail, the jerking motion accelerated by gravity made the TDS service loop cable hit IP # 2, who was standing nearby, resulting in IP # 2 losing his balance and falling approximately 40 feet to the ground. IP 2 sustained a fatal head injury.

WHAT WENT WRONG:

- Inadequate Identification of Worksite Hazard.
- Inadequate Leadership / Supervision.
- In effective Monitoring / Auditing of Safety Process.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- PTW authorities must review the PTW and associated documents for adequacy before signatures.
- Review the existing JSA for rigging down TDS to cover the risk of sudden movement of the TDS and implement the revised JSA on rigs having similar TDS.
- Prepare a consolidated list of rig move activities and enhance the Rig Move on Paper (RMOP).
- Enhance the quality of the Integrated Work Activity Planning (IWAP) process by comprehensively planning and scheduling all activities during rig move and include DTMs in the rig move IWAP meetings.
- Adequate and constant supervision is required to ensure safe execution of the task.

CAUSAL FACTORS:

No Causal Factors Allocated

MIDDLE EAST ONSHORE

DATE: 02 Sep 2022

COUNTRY: UAE

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

Employer: Contractor; **Occupation:** Drilling/Well Servicing Operator; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Unspecified

NARRATIVE:

The operation was dismantling the rig in preparation for a marine move. After disconnecting the monkey board from the mast, the rig crew were moving the monkey board from beneath the mast. The monkey board was resting vertically on the ground on its front side guardrails and a crane's fast line was connected to the monkey board pad eye and kept under tension. The crew were using the crane fast line on off-driller side and two forklifts diagonally. The crane operator slacked the fast line wire completely and as the monkey board was engaged by forklift # 2, the monkey board tipped over onto the floorman (IP), causing a fatal injury.

WHAT WENT WRONG:

- Inadequate Leadership
- Inadequate Planning / Risk Assessment Performed
- Inadequate Communication Between Work Groups
- Inadequate Communication Between Shifts
- Lack of Procedure of the Task
- Inadequate Implementation of Procedure

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Area Authority to ensure application of adequate PTW & associated documents such as JSA and Lift Plan.
- Field Safety Engineers (FSE) to conduct effective Tier-1 audit on daily basis and intervene when required.
- Communicate observations and interventions as part of existing HSE observation system.
- Ensure that lift plans are approved by competent Lifting Specialist.
- Drops zone to be barricaded and safe distance to be maintained from lifting operations.

CAUSAL FACTORS:

No Causal Factors Allocated

MIDDLE EAST OFFSHORE

DATE: 26 Dec 2022

COUNTRY: UAE

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Other

Employer: Contractor; **Occupation:** Foreman, Supervisor; **Body part:** Neck/torso/spine; **Nature of injury:** Fracture

NARRATIVE:

An old hose hangar made of metal frame (weighing approximately 240 kg) was kept in a vertical position at the main deck STBD side area beside a pipe rack. While the service company well testing supervisor was locating a spare elbow in the area, the frame became unbalanced and fell, hitting the IP on his neck, leading to a fatal injury.

WHAT WENT WRONG:

- Improper decision making/lack of judgement by supervisor and chief operator.
- Improper position or posture for the task.
- Improper placement of materials.
- Workplace layout congestion or restricted motion.
- Inadequate Audit/Inspection and Monitoring.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Safety stand-down was conducted across all sites.
- Hazard hunt survey conducted across all sites.
- Removed all unnecessary items, tools or equipment (which are not required for the jobs) from work site immediately.
- Conducted refresher session on Situational Awareness through practical session, with all crew members.
- Hazard hunt conducted with proper responsibilities and accountabilities, involving service company supervisors.
- All to conduct last minute dynamic risk assessment (Step Back 5 x 5) for all jobs/ activities.
- Enforce compliance with WMS, where Area Authority/ Supervisors to physically visit work location prior permit approval.

CAUSAL FACTORS:

No Causal Factors Allocated

MIDDLE EAST OFFSHORE

DATE: 26 Nov 2022

COUNTRY: UAE

NUMBER OF DEATHS: 2

FUNCTION: Production

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Transport - Land

RULE: Driving

Employer: Contractor; **Occupation:** Other; **Body part:** Head (incl. mouth, etc.); **Nature of injury:** Crushing

Employer: Contractor; **Occupation:** Transportation Operator; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Crushing

NARRATIVE:

A vehicle (double cabin pickup) was carrying five occupants when the driver lost control and the vehicle deviated from the road and collided with a 48" fire water line and went through the gap underneath the line. This resulted in fatal injuries to the driver and a passenger, and three passengers sustained serious injuries. Emergency response teams along with medics responded and rescued and transfer casualties to the medical centre. HEMS were activated and the injured were evacuated to hospital. All three injured persons were treated and later discharged from hospital.

WHAT WENT WRONG:

Inadequate Audit/ Inspection/ Monitoring: There was no process in place to detect and correct issues pertaining to fatigue of critical staff such as Drivers and front liners.

Inadequate Work Planning: There was no dedicated road safety coordinator to manage logistic related risks. There was no process in place to monitor welfare of short-term contractors/service providers.

Inadequate Correction of Work Site Job Hazards: IVMS was not effectively configured to capture violations and risk factors.

Inadequate Technical Design: There was a gap under the Fire water line; roadside barriers were not adequate to stop a moving vehicle; the road is close to the Fireline and its associated gap.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Create fatigue awareness sessions for leadership and supervisors.
- Conduct monthly welfare inspections to review crew working hours.
- Assign a road safety Coordinator to review IVMS and develop actions to enhance compliance efforts in promoting road safety.
- Automate timesheets of workers and set triggers intervention for fatigue.

Configure IVMS settings to:

- Generate alarm for over speeding (40 km/Hrs)
- Triggering of SMS/emails for reckless violations, fatigue, and behaviour

Risk asses all routes in industrial area and develop action plan to

- Enhance roadside barriers
- Road Layout
- Road Marking

CAUSAL FACTORS:

No Causal Factors Allocated

MIDDLE EAST OFFSHORE

DATE: 22 Mar 2022

COUNTRY: UAE

NUMBER OF DEATHS: 1

FUNCTION: Construction

CAUSE: Dropped objects

ACTIVITY: Construction, commissioning, decommissioning

RULE: Line of fire

Employer: Contractor; **Occupation:** Manual Labourer; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Unspecified

NARRATIVE:

An electrical helper was pulling out a piece of unistrut channel piled on a scaffolding rack at a laydown area. While pulling out the channel, the scaffolding rack collapsed on him causing fatal head injuries.

WHAT WENT WRONG:

1. Inadequate Leadership:

There was no effective HSE oversight on contractors. The mandate of Operation/Site HSE was limited to Operations areas and Brownfields. Hence the contractor/subcontractors were not subjected to effective inspection/ monitoring.

2. Inadequate audit/ Inspection/ Monitoring:

Contractor HSE had limited interaction with project Site HSE team who oversees plant operations and activities in Brown fields; there was no external structured audit/inspection performed.

3. Inadequate Work Planning:

There was lack of oversight from management resulting in shortage of manpower to perform critical tasks. HSE Manager position remained vacant (filled 3 months back when the Project has completed approximately 60% of the job); Environmental Engineer and Safety Officer positions are still vacant.

4. Inadequate Performance Measurement and Assessment:

Contractor scored 51% in the HSE performance evaluation (July to December 2021) requiring significant improvement.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Enhance/ Streamline/ re-structure HSE resources to provide oversight on all operations and activities (Plant, Brown & Green Field activities) with clear and defined mandates.
- Fill all identified project positions and consider assigning a PMTC HSE Manager on site.
- Enhance weightage of sub-contractor performance in contractors' HSE Assurance Plan to place higher focus on management of sub-contractors.
- Review and align emergency plan with Company Emergency Response Procedure.
- Ensure availability of trained First Aiders and conduct frequent medical emergency drills at all contractor work sites.

CAUSAL FACTORS:

No Causal Factors Allocated

NORTH AMERICA ONSHORE

DATE: 06 Mar 2022

COUNTRY: USA

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

Employer: Contractor; **Occupation:** Process/Equipment Operator; **Body part:** Neck/torso/spine;

Nature of injury: Crushing

NARRATIVE:

Contractor crew was attempting to connect a tractor to a conveyor belt trailer. A contractor worker walked behind the conveyor belt trailer and was caught between the trailer and the blender.

WHAT WENT WRONG:

Personnel did not identify hazards associated with trailer design, which was not standard for service.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Area between trailer and blender considered an exclusion zone during connection.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 20 Sep 2022

COUNTRY: USA

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Drilling, workover, well operations

RULE: Energy isolation

Employer: Contractor; **Occupation:** Other; **Body part:** Head (incl. mouth, etc.); **Nature of injury:** Unspecified

NARRATIVE:

A contractor employee was fatally injured when he was struck by the counterweights while working on a pumpjack isolation.

WHAT WENT WRONG:

Working on equipment which was not energy isolated.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

NORTH AMERICA ONSHORE

DATE: 03 Dec 2022

COUNTRY: USA

NUMBER OF DEATHS: 1

FUNCTION: Unspecified

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

Employer: Contractor; **Occupation:** Drilling/Well Servicing Operator; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Crushing

NARRATIVE:

On December 4, 2022, a floorhand on a drilling rig was struck in the head and shoulder by a fold-up walkway that fell from the tubing board while racking back a stand of pipe. The injured contractor was transported by ambulance to hospital, where the IP was listed in critical condition. The IP passed away as a result of his injuries on 6 December 2022.

WHAT WENT WRONG:

Equipment at height broken off by drawworks. No secondary retention device.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Investigation ongoing.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: 08 Jul 2022

COUNTRY: USA

NUMBER OF DEATHS: 1

FUNCTION: Unspecified

CAUSE: Struck by (not dropped object)

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

Employer: Contractor; **Occupation:** Drilling/Well Servicing Operator; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Crushing

NARRATIVE:

A drilling crew member sustained a fatal injury after being struck by the "grasshopper" electrical delivery mechanism (also called the umbilical cable tray) while rigging down the drilling rig.

WHAT WENT WRONG:

The IP was crushed when cable snapped.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Line of fire must be covered on JSA. Lack of recognition of lifting hazards during rig move.

NORTH AMERICA ONSHORE

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

NORTH AMERICA OFFSHORE

No fatal incidents reported

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 05 Jul 2022

COUNTRY: Kazakhstan

NUMBER OF DEATHS: 2

FUNCTION: Construction

CAUSE: Pressure release

ACTIVITY: Construction, commissioning, decommissioning

RULE: Energy isolation

Employer: Contractor; **Occupation:** Unknown; **Body part:** Internal organs; **Nature of injury:** Crushing

Employer: Contractor; **Occupation:** Unknown; **Body part:** Internal organs; **Nature of injury:** Crushing

NARRATIVE:

An incident occurred resulting in two fatalities and two injuries. A crew of four people were working on the blind flange connection at a 36" line. Another crew had completed a hydrotest on that same line on July 4th. As the crew was working, the blind flange was displaced. The sudden release of pressure caused injuries that resulted on two fatalities and hospitalization of two other employees. A fifth person who was not part of the crew but located in the same working area was seen at the clinic on site and later released.

WHAT WENT WRONG:

Displaced blind flange caused a sudden release of pressure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not yet available.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective Personal Protective Equipment

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 23 Aug 2022

COUNTRY: Kazakhstan

NUMBER OF DEATHS: 1

FUNCTION: Construction

CAUSE: Struck by (not dropped object)

ACTIVITY: Construction, commissioning, decommissioning

RULE: Energy isolation

Employer: Contractor; **Occupation:** Unknown; **Body part:** Head (incl. mouth, etc.); **Nature of injury:** Crushing

NARRATIVE:

A business partner crew completed a pneumatic leak test of the chilled water/return line in the utilities building and then had transitioned to a function test of the flushing equipment. During the flushing equipment function test, one of the temporary 4-inch hoses connected to the chiller system disconnected from its coupling, causing a release of air. Additionally, air was released through a discharge hose that was routed from one of the temporary manifolds to the top of a tank. The hose began to move in an uncontrolled motion as air reached the end of the hose. A crew member evacuating from the chiller room through the western exit was fatally struck by the hose.

WHAT WENT WRONG:

Hose with energy (air) struck person.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not yet available.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

RUSSIA & CENTRAL ASIA OFFSHORE

No fatal incidents reported

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 01 May 2022

COUNTRY: Argentina

NUMBER OF DEATHS: 1

FUNCTION: Exploration

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Seismic/survey operations

RULE: Line of fire

Employer: Contractor; **Occupation:** Heavy Equipment Operator; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Crushing

NARRATIVE:

Working with heavy vibrating equipment, the operator got down to carry out an unforeseen task between equipment, and the equipment that was coming up behind did not see him and when it started it trapped him.

WHAT WENT WRONG:

- WORK FACTORS - INADEQUATE COMMUNICATION - Inadequate communication between peers - There is no established criteria to communicate the need to descend from the vibro or notify the return.
- JOB FACTORS - IMPROPER TOOLS AND EQUIPMENT - Inadequate consideration of human/ergonomic factors - Equipment design does not take into consideration blind spot elimination or descent warning.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Include within the vibro operation process a communication protocol for descending from the vibros.
- Establish technical mechanisms to eliminate blind spots and warn of the presence of people in the vicinity of the equipment.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: 12 Aug 2022

COUNTRY: Argentina

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Excavation, trenching, ground disturbance

RULE: Line of fire

Employer: Contractor; **Occupation:** Other; **Body part:** Respiratory system;

Nature of injury: Major/multiple system trauma

NARRATIVE:

The incident occurred while the victim was working alone excavating materials at a quarry. The victim called his colleagues to inform them the loading shovel was stuck. His colleagues arrived seven minutes later and found that the quarry wall had collapsed, and the operator was partially buried in his machine. They attempted to assist the victim without success. Death was confirmed later by the medic.

SOUTH & CENTRAL AMERICA ONSHORE

WHAT WENT WRONG:

The sand and gravel wall of the quarry collapsed and shifted onto the loading shovel.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Analyse quarry designs with geotechnical specialist.
- Identify areas in quarries where works are not permitted.
- Ensure that the landslide risks associated with quarry operations are analysed.
- Ensure availability of means of communication for activities carried out in isolated areas.
- Ensure that lone worker situations are identified and that their associated risks are adequately managed.
- Ensure equipment are adequate.
- Review personnel competencies.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: 20 Jun 2022

COUNTRY: Argentina

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Production operations

RULE: Line of fire

Employer: Contractor; **Occupation:** Process/Equipment Operator; **Body part:** Head (incl. mouth, etc.);

Nature of injury: Major/multiple system trauma

NARRATIVE:

During test operations with flowback equipment from an exploratory well, while routine tasks were being carried out in the high-pressure sand separator area, an unplanned release of energy occurred.

WHAT WENT WRONG:

- Inadequate risk identification.
- Inadequate change management.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Execution of recommended practice for well de-sanding activities.

CAUSAL FACTORS:

No Causal Factors Allocated

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 08 Nov 2022

COUNTRY: Brazil

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Unspecified - other

RULE: Line of fire

Employer: Contractor; **Occupation:** Heavy Equipment Operator; **Body part:** Leg/knee/hip;

Nature of injury: Cut, Puncture, Scrape

NARRATIVE:

A contracted company employee was walking in an operational area crossing an internal road when he was run over by a forklift. He was promptly rescued but died in the hospital on the same day.

WHAT WENT WRONG:

- RC1: Failure in the forklift preventive driving process.
- RC2: Forklift speed control process failure.
- RC3: Cell phone control failure.
- RC4: Failure in the process of crossing people in the operational area.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review speed limit procedures and review signs in areas.
- Do not use the cell phone in operational areas.
- Implementation of technologies in forklifts – ex: speed limiter, proximity sensor, cameras, alarms, etc.
- Review the obligation to wear hearing protection in operational areas.

CAUSAL FACTORS:

No Causal Factors Allocated

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 02 Aug 2022

COUNTRY: Brazil

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature

ACTIVITY: Maintenance, inspection, testing

RULE: Energy isolation

Employer: Contractor; **Occupation:** Maintenance, Craftsman; **Body part:** Respiratory system;

Nature of injury: Inhalation, poisoning, intoxication, asphyxiation, drowning

NARRATIVE:

Four workers were carrying out maintenance work in the auxiliary motor generator room when the CO₂ fire-fighting system was spuriously activated, flooding the room. Two workers escaped the room in time (without injury/functional disturbance) and the other two workers were rescued by the emergency response team, one of whom died.

WHAT WENT WRONG:

- **CHANGE MANAGEMENT:** failure to apply hazard identification mechanisms and/or risk analysis prior to implementing the modification.
- **WORK ENVIRONMENT AND HUMAN FACTORS:** Requires decision based on knowledge.
- **MECHANICAL INTEGRITY:** There is no manufacturer's recommendation and/or standard and/or standards for inspection, testing and maintenance procedures.
- **MECHANICAL INTEGRITY:** Inspection, testing and maintenance procedure does not have clear instructions for the safe conduct of activities / Incomplete or situation not contemplated.
- **RISK IDENTIFICATION AND ANALYSIS:** Inadequate identification of the necessary actions for risk mitigation and prevention.
- **OPERATIONAL PROCEDURES:** Incomplete Operational Procedure or situation not covered.
- **QUALIFICATION, TRAINING AND PERFORMANCE OF PERSONNEL:** inadequate analysis of training needs for the service and/or tasks.
- **WORK ENVIRONMENT AND HUMAN FACTORS:** Inadequate location of equipment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review of the lockout procedure contemplating the use of the CO₂ system lockout.
- Issuance of technical guidance for the release of work in CO₂ protected environments.
- Review of the attribution matrix for the maintenance process, clearly defining attributions for operation, maintenance, and technical responsibility for the CO₂ system.
- Technical training, with practical emphasis, on the CO₂ systems installed in the units, for professionals who operate, carry out maintenance, testing and inspection.
- Creation of a corporate Working Group to define the Change Management process in automation software in critical systems.

CAUSAL FACTORS:

No Causal Factors Allocated

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 16 Mar 2022

COUNTRY: Brazil

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Aviation accident

ACTIVITY: Transport - Air

RULE: Other

Employer: Contractor; **Occupation:** Other; **Body part:** Internal organs;

Nature of injury: Inhalation, poisoning, intoxication, asphyxiation, drowning

NARRATIVE:

A S76 C++ helicopter collided against the water about 900 metres from an oil rig after an inflight loss of control on the final approach to the helideck. After the impact the helicopter assumed an inverted position. The 11 passengers and the co-pilot managed to escape the inverted helicopter. The captain, although rescued by the co-pilot after his absence on the surface was noticed, did not survive.

WHAT WENT WRONG:

The causes of the accident are being investigated by the Brazilian aeronautical authority.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not yet available.

CAUSAL FACTORS:

No Causal Factors Allocated

For the full analysis of 2022 results,
as well as all fatal incident reports
submitted to IOGP since 1991,
visit IOGP's data website at:
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